Product Requirements Document

Agri Shop Management System — Final Baseline

Prepared as a single source of truth for AI-assisted product design and application development.

Product	Agri Shop Management System
Product Type	Internal Retail / POS + Inventory + Credit & Payables
Primary Users	Shop Owner / Admin and Shop Employees
Initial Categories	Sprays and Seeds
Version	1.0 — Final Baseline
Status	Requirements aligned with finalized discussion

Important implementation note: This document intentionally separates finalized MVP requirements from pending/future architecture so an AI design/build tool can generate the core application without inventing conflicting business rules.

1. Product Overview

We are building an internal management application for a small agricultural retail shop with approximately 3–4 employees. The shop sells agricultural products, initially sprays and seeds, from different companies that are treated as the shop's suppliers/manufacturers.

The application combines retail sales, inventory management, customer management, customer credit, supplier purchases and payables, payment histories, employee sales tracking, product expiry monitoring, stock alerts, and admin-only profitability reporting.

The product should feel like a simple, fast retail POS and inventory system rather than a complex ERP.

2. Product Goals

- Maintain accurate inventory and stock movement.
- Process customer sales quickly.
- Automatically deduct sold quantities from inventory.
- Track customer purchases, payments, credit, and outstanding balances.
- Track supplier purchases, payments, and amounts payable.
- Monitor product expiry and low-stock conditions.
- Track employee sales performance.
- Calculate and report gross profit to Admin.
- Reduce manual calculations and manual record keeping.
- Give the Shop Owner complete operational and financial visibility.

3. Users and Permissions

3.1 Shop Owner / Admin

- Full access to products, inventory, companies/suppliers, purchases, sales, customers, employees, payments, returns, adjustments, reports, and settings.
- Can add and manage products and companies/suppliers.
- Can add stock and record purchases.
- Can manage selling prices and product information.
- Can manage employees and deactivate employee accounts.
- Can view supplier purchase cost, supplier payable, payment history, and shop profit.
- Can view all customer financial information and all employee activity.
- Can manually adjust inventory with a recorded reason.

3.2 Employee

- Primary responsibility: sales and customer operations.
- Can create sales, select products/variants, record quantities and payments, and handle customer-related operations.
- Can view customers, their purchase history, payment history, total purchased amount, total paid, and total outstanding balance.
- All employees have customer visibility across the shop; customer financial history is not restricted to the employee who originally made a sale.
- Cannot add or modify products.

- Cannot change product prices.
- Cannot add/manage companies or suppliers.
- Cannot view supplier financial information.
- Cannot view base/purchase cost or gross profit.
- Cannot manually adjust inventory.
- Cannot perform administrative controls.

4. Company / Supplier Model

Company, supplier, and manufacturer are treated as the same business entity for this product. There is no separate supplier/manufacturer entity in the initial model.

- Company name
- Company logo
- Products supplied
- Purchase history
- Quantities supplied
- Purchase cost
- Total paid
- Outstanding payable
- Supplier payment history

Employees may see company/product information needed for sales, but supplier financial information is Admin-only.

5. Product Management

- Admin creates and manages products.
- Every product belongs to a company/supplier.
- Initial categories: Spray and Seed.
- The category model should be extensible for future agricultural categories.
- Products must be searchable and filterable by category and company/supplier.
- Product details must support the appropriate measurement/pricing model for its category.

6. Product Units, Variants and Pricing

6.1 Spray

Sprays are sold as pack-size variants. Inventory is tracked separately for each sellable variant.

- Examples: 1 Liter, 500 ml, 200 ml.
- Each variant has its own sellable quantity and current selling price.
- Example: Janja Spray — 1L: 20 units; 500ml: 35 units; 200ml: 10 units.
- Selling 3 × 500ml reduces only the 500ml stock.

The system may calculate proportional prices from a base pack size, but Admin must be able to define/override the actual selling price for a variant. The business does not require smaller spray packs to always have mathematically proportional retail pricing.

6.2 Seed

Seeds use a fixed price-per-kilogram model.

- Example: Rs. 1,000 per kg.
- 10 kg = Rs. 10,000.
- 500 g = Rs. 500.
- The system calculates the sale amount from quantity × configured per-KG rate.
- Seed pricing should not be treated like arbitrary spray pack variants.

7. Inventory Management

- Admin adds stock through a stock-in/purchase workflow.
- Inventory is tracked at the product/variant/measurement level.
- Inventory records quantity, base/purchase price, selling price, expiry date, and stock status.
- Completed sales automatically reduce inventory.
- Approved returns automatically increase inventory.
- Admin stock adjustments update inventory and create a permanent adjustment record.
- Inventory cannot become negative.

7.1 Stock Status

- In Stock
- Limited Stock
- Low Stock
- Out of Stock
- Expired
- Not Sellable

Low-stock thresholds should be configurable. Example: 100 units initially, 90 sold, 10 remaining; if the configured threshold is 10, the item becomes Low Stock. Low-stock/expiry warnings should primarily be visible in Inventory rather than being repeated across every screen.

7.2 Expiry

- Applicable products have an expiry date.
- Inventory should surface products approaching expiry and expired products.
- Expired stock remains represented for accurate reconciliation but is marked Expired / Not Sellable.
- Expired stock must not be available for normal sale.
- The expiry warning window should be configurable.

7.3 Batch-wise expiry — Pending / Future

The same product/variant may eventually have multiple batches with different expiry dates. This is intentionally pending for a future phase. The architecture should remain extensible to support batch-level stock and expiry later without rebuilding the core product.

8. Stock Purchase / Stock-In

Flow: Admin → Purchase / Stock In → Select Company/Supplier → Select Product → Select Unit/Variant → Enter Quantity → Enter Base/Purchase Price → Enter Selling Price → Record Payment → Save.

The system adds the purchased quantity to inventory and separately records the supplier's financial position.

Example: 100 units × Rs. 2,000 base price = Rs. 200,000 purchase cost. If Rs. 100,000 is paid, inventory still receives all 100 units and supplier payable becomes Rs. 100,000.

9. Supplier Payables and Payments

Supplier financial tracking mirrors the customer-credit concept in the opposite direction: the shop owes the supplier when a purchase is unpaid.

- Existing payable + new unpaid purchase = new cumulative payable.
- Supplier payments reduce payable.
- Every supplier payment is permanently recorded.
- Supplier history should show purchases, amounts, payments, and remaining payable.

Flow: Supplier → Outstanding → Record Payment → Payable Reduced → Permanent Payment History.

Example: Previous payable Rs. 300,000; payment Rs. 50,000; remaining payable Rs. 250,000.

10. Customer Management and Credit

- Customer profile includes name, phone number, address, purchase history, payment history, total purchases, total paid, outstanding amount, and credit limit.
- All employees can access the customer's complete shop-wide financial history.
- Employees cannot access Admin-only supplier cost/profit information.
- Customer can have multiple outstanding sales/credit transactions.
- Customer-level outstanding is the cumulative unpaid amount.

10.1 Payment Types

- Full Payment: total paid equals total sale amount; outstanding is zero.
- Partial Payment / Credit: amount paid is less than total; system calculates the outstanding amount automatically.

10.2 Credit Limit

Admin can define a credit limit for each customer. If a new credit transaction would exceed the limit, the system should warn the user and apply the configured permission/business rule.

10.3 Credit Settlement

A later customer payment creates a new permanent payment record and reduces outstanding balance. The system must not overwrite historical transactions.

11. Customer Financial History

Customer history should preserve the full transaction trail. Each sale/payment context should allow the user to understand what was purchased, how much it cost, how much was paid, what remained outstanding, when it occurred, and which employee processed the original sale.

When a new payment is made, it should create a new transaction/history record and update the customer's current balance.

12. Sales Management

Flow: Employee → Select Customer → Select Product → Select Variant/Unit → Enter Quantity → Review Sale → Select Payment Type → Enter Amount Paid → Complete Sale.

- A sale may contain multiple products.
- System calculates line totals and overall total automatically.
- System calculates amount paid and outstanding automatically.
- Every sale records customer, employee, products, quantities, prices, payment information, and date/time.
- Completed sale automatically updates inventory and customer financial state.

12.1 Historical Price Rule

Historical sales retain the selling price that was effective at the time of the sale. A later selling-price change must never recalculate old sales.

13. Price and Cost Rules

- Base/purchase price is Admin-only.
- Employees never see purchase cost.
- Existing stock keeps its recorded base/purchase cost.
- The base cost is not retroactively changed for existing stock.
- A new stock purchase may have a different base cost.
- Selling price can change over time by effective date.
- Remaining inventory uses the current applicable selling price.
- Historical sales preserve their original selling price.

Example: old stock cost Rs. 4,000 and sold at Rs. 5,000; later selling price becomes Rs. 5,500. Existing historical sales remain at Rs. 5,000. New sales use Rs. 5,500. A future stock purchase can have a different base cost.

14. Gross Profit and Admin Reporting

Gross profit is Admin-only.

Gross Profit = Sales Revenue − Cost of Goods Sold

- Today's gross profit
- Yesterday's gross profit
- This week's gross profit
- This month's gross profit
- Previous month
- Multiple-month periods
- Six-month period
- One-year period
- Custom date range

Admin should be able to drill into sales by employee, company/supplier, product, category, customer, payment status, and date range. Example: Employee A → Janja → August → Spray.

15. Returns and Refunds

Returns should originate from an original completed sale. An employee must not be able to create an arbitrary return for a product that the customer did not purchase.

Flow: Original Sale → Select Product → Return Quantity → Verify Customer/Transaction → Choose Refund or Credit Adjustment → Confirm.

- Return quantity cannot exceed the quantity originally sold/eligible for return.
- Inventory is increased by the approved returned quantity.
- Customer financial balance is updated automatically.
- Refund option returns money to the customer.
- Credit Adjustment option applies the return value against the customer's outstanding balance.
- Original sale remains intact; the return is recorded as a separate transaction.

Basic return functionality is preferred for MVP if implementation remains manageable. Advanced exchange/complex refund workflows can be future scope.

16. Inventory Adjustment

Admin can manually adjust stock only through a recorded adjustment workflow.

- Damaged
- Missing
- Expired
- Manual Correction
- Other

Each adjustment must record the quantity change, reason, date/time, and responsible Admin/user. The system must not silently change stock.

17. Employee Dashboard

Employee dashboard is operational, not financial-profit focused.

- Today's total sales
- Number of transactions
- Customers served
- Products sold
- Quantities sold
- Paid sales
- Credit sales
- Relevant outstanding credit/customer activity
- Sales history

Employee dashboard supports Today, Yesterday, This Week, This Month, and Custom Range filters. Employee can access customer details and complete customer history but cannot access supplier costs, supplier payable, or profit.

18. Admin Dashboard

Admin dashboard provides complete shop-wide visibility. Default period is Today, with date/range controls.

- Total sales
- Number of transactions
- Paid sales
- Credit sales
- Outstanding customer credit
- Gross profit
- Employee-wise sales
- Company/supplier-wise sales
- Product/category performance
- Customer activity
- Low-stock products
- Out-of-stock products
- Expiring/expired products
- Supplier payable

Recommended filters: Date/date range, Employee, Company/Supplier, Product, Category, Customer, Payment Status, Credit Status.

19. Company / Supplier Detail

Admin can open a company/supplier and see:

- Company name and logo
- Products supplied
- Purchase history
- Quantities purchased
- Purchase/base cost
- Total purchased amount
- Total paid
- Current payable
- Supplier payment history
- Related inventory history

Supplier financial information is Admin-only.

20. Employee Management

- Admin can create employees.
- Admin can edit employee details.
- Admin can manage access.
- Admin can deactivate employees.
- Employee accounts should not be hard-deleted when they leave.
- Historical sales and transactions remain connected to the inactive employee.
- Inactive employees remain available in historical reporting.

21. Record Integrity and Deletion

Completed financial, sales, purchase, payment, return, and inventory movement records should not be hard-deleted.

- Sales: preserve history.
- Payments: preserve history.
- Purchases: preserve history.
- Stock movements: preserve history.
- Returns: preserve history.
- Adjustments: preserve history.
- Corrections should use cancellation, reversal, return, or adjustment mechanisms.

22. Reporting and Filters

Admin reporting should provide full visibility without requiring the user to manually cross-reference records.

- Date / custom date range
- Employee
- Company/Supplier
- Product
- Category
- Customer
- Payment status
- Credit status

Employees may use the date-range filters for their own sales/customer activity, subject to their role permissions.

23. Information Architecture

Recommended Admin navigation:

- Dashboard
- Sales
- Customers
- Products
- Inventory
- Companies / Suppliers
- Purchases / Stock In
- Employees
- Payments
- Reports
- Settings

Recommended Employee navigation:

- Dashboard
- New Sale
- Sales
- Customers
- Products

The final navigation can be optimized during UX design as long as role permissions and core workflows remain intact.

24. Core Entity Relationships

Company/Supplier → Product → Product Variant/Unit → Inventory

Company/Supplier → Purchase/Stock-In → Supplier Payable → Supplier Payment

Customer → Sale → Payment → Customer Outstanding

Employee → Sale → Customer

Product/Inventory → Sale → Stock Deduction

Original Sale → Return → Inventory Adjustment + Customer Financial Adjustment

25. Core Business Rules

- Selling a product automatically reduces the relevant inventory quantity.
- Returning a product automatically increases the relevant inventory quantity.
- Stock cannot become negative.
- Low-stock status is threshold-based and primarily visible in Inventory.
- Expired products are Not Sellable.
- Every sale is associated with a customer and employee.
- Every sale preserves the price effective at the time of sale.
- Customer outstanding is cumulative across unpaid transactions.
- Customer payments are permanent records.
- Supplier payable is cumulative across unpaid purchases.
- Supplier payments are permanent records.
- Admin can see purchase costs and profit; employees cannot.
- Admin can adjust inventory; employees cannot.
- Employee deactivation does not remove historical records.
- Completed records are not hard-deleted.

26. MVP Scope

Required for MVP

- Admin and Employee authentication
- Role-based permissions
- Employee creation/deactivation
- Company/Supplier management

- Product management
- Spray variants/pack sizes
- Seed per-KG pricing
- Base/purchase price
- Selling price
- Expiry date
- Inventory and stock-in
- Automatic stock deduction
- Low-stock status
- Out-of-stock status
- Expired/not-sellable status
- Stock adjustments
- Customer management
- Customer purchase history
- Customer payment history
- Customer credit
- Customer credit limit
- Supplier payable
- Supplier payments and payment history
- Sales creation
- Full and partial payment
- Employee sales tracking
- Admin sales analytics
- Gross profit reporting
- Date and business filters
- Basic returns if implementation remains manageable

27. Future / Pending

- Batch-level inventory where the same product/variant has different expiry dates.
- Advanced batch-level costing.
- Advanced return/exchange workflows.
- Additional product categories beyond Spray and Seed.
- More advanced supplier/vendor capabilities.
- Any additional functionality that is not required for the core retail loop.

28. UX / Product Design Principles

- Fast sales entry is the highest employee UX priority.
- Use clear retail/POS patterns rather than ERP-heavy workflows.
- Automate calculations wherever possible.

- Keep Admin financial information separate from Employee views.
- Make inventory alerts clear but avoid unnecessary global notification noise.
- Allow drill-down from summary metrics to underlying records.
- Keep transaction history traceable and auditable.
- Use clear statuses for paid, partial/credit, low stock, out of stock, expired, and payable/outstanding states.
- Design the system so future batch/expiry support can be introduced without redesigning the core data model.

29. End-to-End Business Flows

Admin — Add Company

Flow: Companies → Add Company → Name + Logo → Save

Admin — Add Product

Flow: Products → Add Product → Company → Category → Unit/Variant → Base Price → Selling Price → Initial/Stock Details → Expiry → Save

Admin — Purchase / Stock-In

Flow: Purchases/Stock In → Company → Product → Unit/Variant → Quantity → Base Price → Selling Price → Payment → Save → Inventory Increased + Supplier Payable Updated

Employee — Sale

Flow: New Sale → Customer → Product → Variant/Unit → Quantity → Review → Payment → Complete → Inventory Reduced + Customer Balance Updated + Employee Sale Recorded

Customer — Credit Payment

Flow: Customer → Outstanding → Record Payment → Balance Reduced → Permanent Payment History

Supplier — Payment

Flow: Company/Supplier → Payable → Record Payment → Payable Reduced → Permanent Payment History

Return

Flow: Original Sale → Product → Return Quantity → Refund OR Credit Adjustment → Confirm → Inventory Increased + Customer Balance Updated + Return Recorded

Inventory Adjustment

Flow: Inventory → Product → Adjust Stock → Reason → Confirm → Permanent Stock Movement

30. Success Criteria

The shop should be able to operate its core business without separate manual records for inventory, sales, customer purchases, customer credit, customer payments, supplier purchases, supplier payments, supplier payables, expiry monitoring, low-stock monitoring, employee performance, and gross profit.

The Admin should be able to answer: What did we sell today? Who sold it? Who bought it? What products were sold? How much was paid? How much customer credit remains? How much do we owe suppliers? What stock is running low? What products are expiring? How much gross profit was generated? Which employees and companies/products are performing best?

31. Final Product Definition

The final product is a role-based Agri Retail Management System connecting Companies/Suppliers, Products, Product Variants/Units, Inventory, Purchases, Sales, Customers, Payments, Customer Credit, Supplier Payables, Profit, and Employee Performance with complete historical transaction tracking.

The Shop Owner receives complete business visibility and control. Employees receive a fast and simple sales/customer-management experience. The system should be simple enough for daily operation in a small shop while remaining extensible for future batch-level inventory, advanced expiry management, additional categories, and more sophisticated workflows.

32. Requirements Locked for AI Design / Build

The following decisions are explicitly finalized and should not be inferred differently by an AI design/build tool:

- Company = Supplier = Manufacturer in the initial data model.
- Initial product categories are Spray and Seed.
- Sprays use pack-size variants such as 1L, 500ml, and 200ml.
- Seed pricing is based on a fixed per-KG rate.
- Inventory is tracked by sellable product/variant/measurement.
- Admin adds and manages products and stock.
- Employees do not manage products, prices, suppliers, stock adjustments, purchase cost, or profit.
- Customer financial visibility is shop-wide for employees, but Admin-only sensitive business data remains hidden.
- Customer credit limit is configurable by Admin.
- Supplier payable is tracked similarly to customer outstanding credit.
- Supplier and customer payment histories are permanent transaction histories.
- Gross profit is Admin-only and supports historical/custom reporting.
- Selling price changes apply to current/future sales while historical sales retain their original price.
- Existing stock's recorded base cost is not retroactively changed; new stock can have a different base cost.
- Returns originate from original sales and can result in a refund or credit adjustment.
- Inventory adjustments are Admin-only and require a reason.
- Expired stock is retained for reconciliation but is not sellable.
- Employee accounts are deactivated rather than deleted; historical records remain intact.
- Financial and inventory records are not hard-deleted.
- Batch-wise expiry is explicitly pending/future and should not complicate the initial MVP.

33. AI Design / Build Instruction

Use this PRD as the source of truth for designing the application. Derive the necessary screens, navigation, components, forms, states, tables, dashboards, detail views, confirmations, empty states, loading states, validation states, success/error states, and responsive layouts from the requirements above.

Do not invent business rules that conflict with this document. Where the PRD intentionally leaves a UI detail open, choose the simplest professional retail/POS UX that preserves the defined roles, financial relationships, inventory behavior, historical records, and permissions.

Prioritize the complete end-to-end flows over isolated screens. The final design should make it possible to move from dashboard summaries into the underlying sale, customer, product, inventory, employee, company/supplier, payment, and historical records.

Keep the MVP operationally simple while preserving an extensible architecture for pending batch-level expiry and future product categories.